

GSEB Class 10 Science

SECTION C - CHAPTER 1

Chemical Reactions and Equations

3-mark questions | Point-to-point | Exam-oriented

1. Observe the figure and answer the following:

a) What is formed from quick lime and water? Write the reaction.

- Quick lime is calcium oxide (CaO).
- It reacts vigorously with water.
- Calcium hydroxide, called slaked lime, is formed.
- A large amount of heat is released.
- Therefore, the reaction is exothermic and also a combination reaction.
- Balanced equation: $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{heat}$.

b) What is the reaction occurring between quick lime and water called?

- The reaction is called slaking of lime.
- Quick lime means calcium oxide (CaO).
- It reacts rapidly with water.
- Slaked lime, calcium hydroxide, is produced.
- A considerable amount of heat is evolved.
- Hence, it is an exothermic combination reaction.

c) State the chemical formula of slaked lime.

- Slaked lime is calcium hydroxide.
- Calcium hydroxide is formed when quick lime reacts with water.
- Its common name is slaked lime.
- Its chemical name is calcium hydroxide.
- Calcium is represented by Ca and hydroxide by OH.
- Therefore, the chemical formula of slaked lime is Ca(OH)_2 .

2. What is meant by decomposition reaction? Give example.

- A decomposition reaction is a reaction in which a single reactant breaks down into two or more simpler products.
- Decomposition may occur by heat, light or electricity.
- Thermal decomposition occurs on heating.
- Photolytic decomposition occurs in the presence of light.
- Electrolytic decomposition occurs by passing electricity.
- Example: $\text{CaCO}_3\text{(s)} \xrightarrow{\text{heat}} \text{CaO(s)} + \text{CO}_2\text{(g)}$.
- Thus, calcium carbonate decomposes to give calcium oxide and carbon dioxide.

3. Observe the figure and answer the following:

a) State the colour of sodium sulphate and barium chloride.

- Sodium sulphate solution is colourless.
- Barium chloride solution is also colourless.
- Both are aqueous salt solutions.
- No characteristic colour is observed before mixing.
- After mixing, a white solid is formed.
- The white solid is barium sulphate (BaSO_4).

b) Write a balanced chemical equation for the reaction between barium chloride and sodium sulphate.

- The reactants are aqueous barium chloride and aqueous sodium sulphate.
- Barium ions combine with sulphate ions.
- A white insoluble precipitate of barium sulphate is formed.
- Sodium chloride remains in solution.
- The equation must have equal atoms on both sides.
- Balanced equation: $\text{BaCl}_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})\downarrow + 2\text{NaCl}(\text{aq})$.

c) What type of chemical reaction takes place between barium chloride and sodium sulphate?

- The reaction is a double displacement reaction.
- Two compounds exchange their ions.
- Ba^{2+} combines with SO_4^{2-} to form BaSO_4 .
- Na^+ combines with Cl^- to form NaCl .
- Barium sulphate is insoluble in water.
- Therefore, a white precipitate is formed, so it is also a precipitation reaction.

4. What is meant by oxidation and reduction reactions? Explain by giving examples.

- Oxidation is the addition of oxygen or removal of hydrogen from a substance.
- Oxidation can also be described as loss of electrons.
- Reduction is the removal of oxygen or addition of hydrogen.
- Reduction can also be described as gain of electrons.
- Example: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$.
- In this reaction, CuO loses oxygen and is reduced to Cu .
- At the same time, H_2 gains oxygen and is oxidised to H_2O .

5. Ashokbhai started a small-scale industry for potato chips. After some time the product became rancid and its smell and taste changed. What do you understand by rancidity? Also state the remedies to prevent it.

- Rancidity is the oxidation of fats and oils in food when exposed to air.
- It causes an unpleasant smell and change in taste.
- Food can be protected by storing it in airtight containers.
- Refrigeration slows down oxidation and helps prevent rancidity.
- Antioxidants can be added to food to prevent oxidation.
- Potato-chip packets may be flushed with nitrogen gas before sealing.
- These methods reduce contact with oxygen and delay rancidity.

6. Balance the following chemical equations.

- (i) $2\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
- (ii) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
- (iii) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$
- In (i), two NaOH are required to balance Na and OH .
- Equations (ii) and (iii) are balanced as written above.
- Check that the number of atoms of each element is equal on both sides.

7. Hitarthi saw her grandmother storing pickles in ceramic pots. She also knew that pickles should not be stored in metal containers. Her friend often brought pickles wrapped in aluminium foil. Answer the following:

a) Why should pickles not be stored in metal containers?

- Pickles are acidic in nature.
- The acids present in pickles can react with metals.
- Such reactions may form metal salts.
- Hydrogen gas may also be evolved in some metal-acid reactions.
- This can spoil the food and may damage the container.
- Therefore, ceramic or glass containers are preferred for storing pickles.

b) Which substances present in pickles react with the metal containers?

- Acids present in pickles react with metals.
- They include organic acids present in sour ingredients.
- The acidic substances react with reactive metals.
- The reaction can produce salts.
- In suitable cases, hydrogen gas can also be released.
- Therefore, the acids in pickles are responsible for the reaction with metal containers.

c) What value of Hitarthi is seen in the above act?

- Hitarthi shows scientific awareness.
- She applies scientific knowledge to daily life.
- She identifies a possible harmful chemical reaction.
- She gives useful safety advice to her friend.
- She shows care and concern for others.
- Her action reflects a scientific attitude and sense of responsibility.

8. What is called a redox reaction? Explain it with suitable examples.

- A redox reaction is a chemical reaction in which oxidation and reduction occur simultaneously.
- Oxidation means loss of electrons, while reduction means gain of electrons.
- Example: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$.
- Here zinc loses electrons and is oxidised to Zn^{2+} .
- Cu^{2+} gains electrons and is reduced to copper metal.
- Another example is $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, where CuO is reduced and H_2 is oxidised.
- Thus, both processes occur together in a redox reaction.

9. Identify the types of chemical reactions from the following equations:

- (i) $\text{Pb} + \text{CuCl}_2 \rightarrow \text{PbCl}_2 + \text{Cu}$ - Displacement reaction.
- Lead displaces copper from copper chloride.
- (ii) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ - Combustion reaction.
- Methane burns in oxygen to produce carbon dioxide and water.
- (iii) $\text{CaCO}_3 \xrightarrow{\text{heat}} \text{CaO} + \text{CO}_2$ - Decomposition reaction.
- Calcium carbonate breaks down into simpler substances on heating.
- Thus, the three types are displacement, combustion and decomposition.

10. Give the balanced chemical equations for the following reactions.

- (i) Reaction useful in black-and-white photography: $2\text{AgBr(s)} \xrightarrow{\text{sunlight}} 2\text{Ag(s)} + \text{Br}_2\text{(g)}$.
- (ii) Oxidation of glucose: $\text{C}_6\text{H}_{12}\text{O}_6\text{(s)} + 6\text{O}_2\text{(g)} \rightarrow 6\text{CO}_2\text{(g)} + 6\text{H}_2\text{O(l)} + \text{energy}$.
- (iii) Formation of water from H_2 and O_2 : $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$.
- All three equations are balanced.